HIERARCHICAL BAYESIAN INFERENCE OF ROCKFILL STRENGTH PARAMETERS FROM LARGE-SCALE TRIAXIAL TEST DATA

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ABSTRACT

The shear strength of rockfill is often estimated using the empirical Barton-Kjærnsli (B-K) criterion to address testing limitations associated with the large particle size of these materials. In this criterion, the equivalent roughness (R) and strength (S) are the stress-dependent structural component parameters added to the basic friction angle (φ_b) of the parent rock to determine the shear strength of the rockfill material. However, for materials intended for use in tailings dam construction, the strength parameters must be supported by triaxial testing results. Due to the characteristics of the materials, large-scale tests are often limited in number. This challenge is compounded by the heterogeneity of the materials and differing proportions of rock types used in a project area. In previous papers (Linero-Molina et al., 2022; Contreras et al., 2022), the authors discussed the application of a Bayesian approach for inferring B-K parameters from large-scale laboratory testing results. This paper discusses the advantages of applying a hierarchical Bayesian model for optimal estimation of B-K parameters from large-scale triaxial test results, particularly when multiple rockfill mix proportions and lithological types are involved. The Bayesian framework integrates the B-K model, laboratory data from large-scale triaxial and basic friction angle testing, and allowable parameter ranges (priors) to create a posterior probability function. This posterior is evaluated using the Markov Chain Monte Carlo (MCMC) method to estimate the most likely parameter values that align with observed material behaviour. This approach provides insights into parameter correlations and uncertainty, thereby streamlining parameter estimation in an objective and rational manner. The methodology is demonstrated through a case study in which four rockfill material groups comprising mixtures of five rock types are investigated using the available laboratory testing data.

1 INTRODUCTION

Determining the shear strength of rockfill materials is challenging due to the large size of the particles making these materials, which precludes the characterisation based on results from laboratory testing with conventional equipment. The strength characterisation is typically carried out using the empirical Barton-Kjærnsli (B-K) criterion based on adding to the basic friction angle of the parent rock φ_b, the stress-dependent structural component parameters R and S representing the equivalent roughness and strength terms, respectively. While φ_b can be directly evaluated in the laboratory, R and S are typically determined from the empirical B-K charts intended for preliminary estimations, based on conventional assessments of the material. However, for detailed engineering of high-risk category dams, the strength characterisation is frequently based on ad hoc large-scale tests. These tests, however, are often limited in number due to the inherent difficulties of large-scale sampling and testing. This situation is aggravated by the fact that rockfill materials are heterogeneous, occurring with different proportions of the rock types available in a project site depending on the geology and mining plan. Therefore, a limited number of large-scale tests are commonly available to characterise the various material mixtures planned in a project site.

The Bayesian approach for the inference of parameters has several advantages over conventional regression methods. This approach provides a formal basis for integrating multiple sources of information, such as the results from various types of laboratory tests and engineering judgments. It also offers a robust framework for quantifying uncertainty, is capable of handling small sample sizes, and allows for the updating of parameters as new information becomes available. A specific type of Bayesian model, known as hierarchical or multi-level models, additionally provides a structured way to incorporate related but not identical information (Williams and Ledesma, 2022). For instance, the results of triaxial tests on samples with different proportions of rock types at a project site can be included as related information, though they should not carry the same weight as results from a specific material mixture of interest.

The Bayesian approach integrates three key components: prior information about the parameters, the B-K model to predict strength properties of a given rockfill mix, and triaxial test data containing actual rockfill behaviour measurements. These elements are combined into a probabilistic function—the posterior probability distribution—within the Bayesian framework. This posterior provides the probability of specific parameter sets and the objective of the analysis is to define the sets of parameters with higher probabilities, which correspond to the minimum differences between model predictions and observed data. The parameters are inferred by evaluating the posterior using the Markov Chain Monte Carlo (MCMC) method (Kruschke, 2015).

This paper aims to present the benefits of the Bayesian approach for parameter inference and to illustrate the advantages of hierarchical Bayesian analysis compared to independent Bayesian analysis when handling multigroup data, especially with limited data. The paper is organized as follows: Section 2 provides a concise description of the B-K model and discusses its relevance for the Bayesian analysis. Section 3 introduces the Bayesian framework, presenting the generic formulation for the independent analysis of material groups and extending the concept to the hierarchical analysis used for the efficient use of the available data to characterise different material groups. Section 4 presents a case study to demonstrate the application of the proposed method, highlighting the benefits of the hierarchical analysis. Section 5 discusses the findings, and Section 6 presents a summary and final remarks.

2 THE BARTON-KJÆRNSLI STRENGTH MODEL

The evaluation of the strength of a rockfill material is generally based on the Barton-Kjærnsli (B-K) strength model as mentioned previously (Barton and Kjærnsli, 1981), which takes the non-linearity of the shear strength of very coarse granular materials into consideration. The model defines the friction angle of a coarse granular material φ according to Equation 1, where φ_b is the basic friction angle of the rock measured on saw-cut surfaces, R is the equivalent roughness, S is the size-dependent equivalent strength of waste particles and σ_n is the effective normal stress.

$$\varphi = \varphi_b + R\, Log_{10}\left(\frac{S}{\sigma_n}\right) \tag{1}$$

The B-K strength model was developed for use alongside empirical charts that provide estimates of the parameters R and S, which are challenging to define through conventional laboratory tests. The particle shape characteristics and material porosity estimates are used to determine R, while the intact rock strength and characteristic particle size are related to the parameter S. Estimates of the basic friction angle are supported by laboratory measurements on polished surfaces, as it is a conventional and cost-effective test. However, for applications requiring a high confidence determination of rockfill strength properties, such as high-risk category tailings storage dams, the B-K parameters should be validated by laboratory testing results (Hawley, 2017).

The geometrical interpretation of the B-K parameters is illustrated in Figure 1. The graph demonstrates that a single B-K strength envelope can be generated using multiple sets of φ_b, R, and S parameters, fixing one parameter determines the value of the other two. This underscores the significance of φ_b within the model, since by fixing φ_b, the values of R and S are determined for a specific envelope. Consequently, fitting B-K envelopes to the results of large-scale triaxial tests and basic friction angle determinations provides an effective method for defining the strength characteristics of rockfill materials in a rational manner.

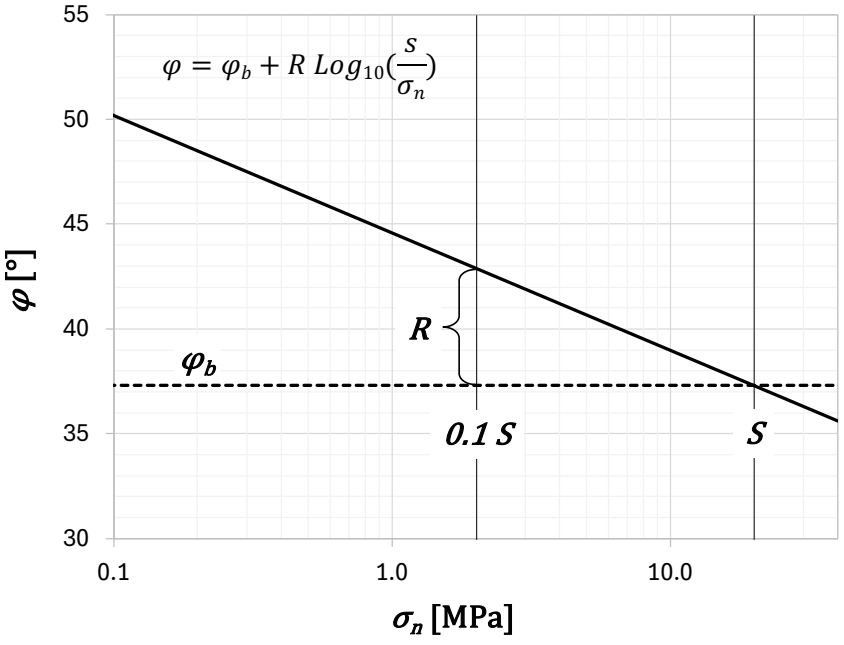


Figure 1: Meaning of the Barton-Kjærnsli strength model parameters

3 BAYESIAN INFERENCE OF THE B-K PARAMETERS

3.1 OVERVIEW OF THE APPROACH

The advantages of the Bayesian approach for inference of parameters are described in detail by Contreras et al. (2018) within the context of intact rock strength. The use of a Bayesian approach for the inference of the B-K model parameters for characterisation of a rockfill material was discussed by Linero-Molina et al. (2022) and Contreras et al. (2022). The Bayesian analysis enables the estimation of the more likely parameter values given three elements as conceptually illustrated in the diagram of Figure 2. First, a model, second, the data to feed the model and third, any prior information on the parameter values.

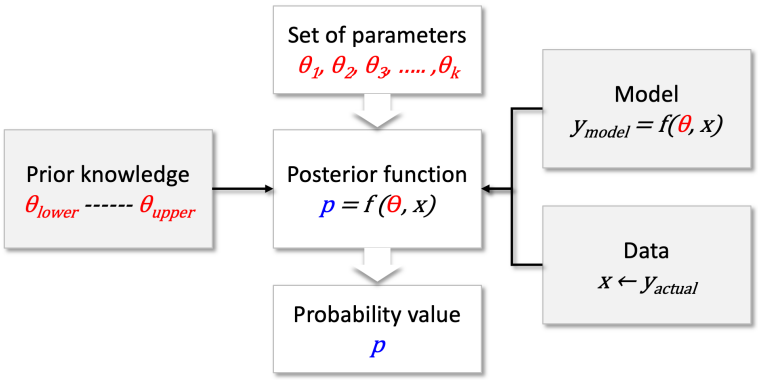


Figure 2: Conceptual representation of the Bayesian model for parameter inference

The model corresponds to a mathematical function f predicting the response y_{model}, of a system of interest.

$$y_{model} = f(\theta, r) \tag{2}$$

In this case, θ and r, are vectors of uncertain and certain model parameters, respectively. Certain parameters include predictor variables x, which define the target response variable y. The data corresponds to measurements of the actual system response, y_{actual}, which are compared with the model predictions to define errors ϵ, representing model uncertainty.

$$\epsilon = y_{actual} - f(\theta, r) \tag{3}$$

The error, ϵ, is assumed to have a normal (Gaussian) distribution around the model prediction, with standard deviation, σ. The standard deviations of the errors are also uncertain parameters that need to be inferred together with the model parameters of interest in vector, θ, and are called nuisance parameters.

The prior information typically consists of valid ranges of credible values of the uncertain parameters to be inferred. These three elements (model, data and prior information) are combined in a probabilistic function that contains the set of parameters for inference. This function corresponds to the posterior probability distribution of the uncertain parameters θ and σ given the data D, and is evaluated using Bayes' theorem:

$$p(\theta, \sigma | D) = \frac{p(\theta, \sigma) p(D | \theta, \sigma)}{p(D)} \tag{4}$$

This function gives probability values for particular sets of parameters and is often expressed as (Kruschke, 2015):

$$posterior = \frac{prior \times likelyhood}{evidence} \tag{5}$$

The prior $p(\theta,\sigma)$ represents initial probabilities for σ and the k uncertain parameters in θ, and is commonly defined with uniform distributions:

$$p(\theta, \sigma) = \frac{1}{\sigma_{upper} - \sigma_{lower}} \prod_{j=1}^{k} \frac{1}{\theta_{upper\ j} - \theta_{lower\ j}} \tag{6}$$

The subscripts in this equation refer to upper and lower values defining credible ranges of the uncertain parameters. The likelihood $p(D|\theta,\sigma)$ is the probability of the data given the uncertain parameters and is defined using a normal distribution to reflect the Gaussian distribution of ϵ. The calculation is carried out for the n measurements of the system response:

$$p(D|\theta,\sigma) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} e^{-\left(\frac{\epsilon^2}{2\sigma^2}\right)} \tag{7}$$

The denominator in Bayes' equation, *p(D)*, is the normalization factor computed as the integral of the numerator over the parameter space. For practical parameter inference using MCMC, this normalization term is not required, and the posterior reduces to:

$$p(\theta,\sigma|D) \propto p(\theta,\sigma)\, p(D|\theta,\sigma) \tag{8}$$

The objective of the analysis is to define the sets of parameters that produce the highest probability values (minimum errors), which can be done with the efficient sampling of the posterior distribution using the MCMC method.

3.2 APPLICATION OF THE APPROACH TO THE B-K MODEL

For the Bayesian evaluation of rockfill strength using triaxial data, the model corresponds to the B-K strength model represented by Eq. 1. In this case, φ is the predicted variable to be compared with the data values, σ_n is the predictor variable that points to the predicted variable within the model and φ_b, R and S are the uncertain parameters to be inferred with the Bayesian approach to minimise the errors, i.e. the differences between the model predictions and the data measurements. The data consist of large-scale triaxial strength test results consisting of measurements of friction angle (φ) for various levels of effective normal stress (σ_n). The prior information consists of the known ranges of credible values for the uncertain parameters in the B-K equation.

The diagram in Figure 3 shows the details of how the Bayesian model components are integrated to define the posterior function. The errors are evaluated with a normal distribution to construct the likelihood function. The method considers prior information distributions on four parameters: φ_b, R, S and the standard deviation of the normal distribution σ_φ, used to model the errors. The specification of the prior information distributions is based on the characteristics of the datasets and consists of setting up values sufficiently vague to avoid constraining the result. The vague priors of the parameters are represented by uniform distributions. The posterior probability distribution is evaluated by sampling the parameters with an MCMC algorithm.

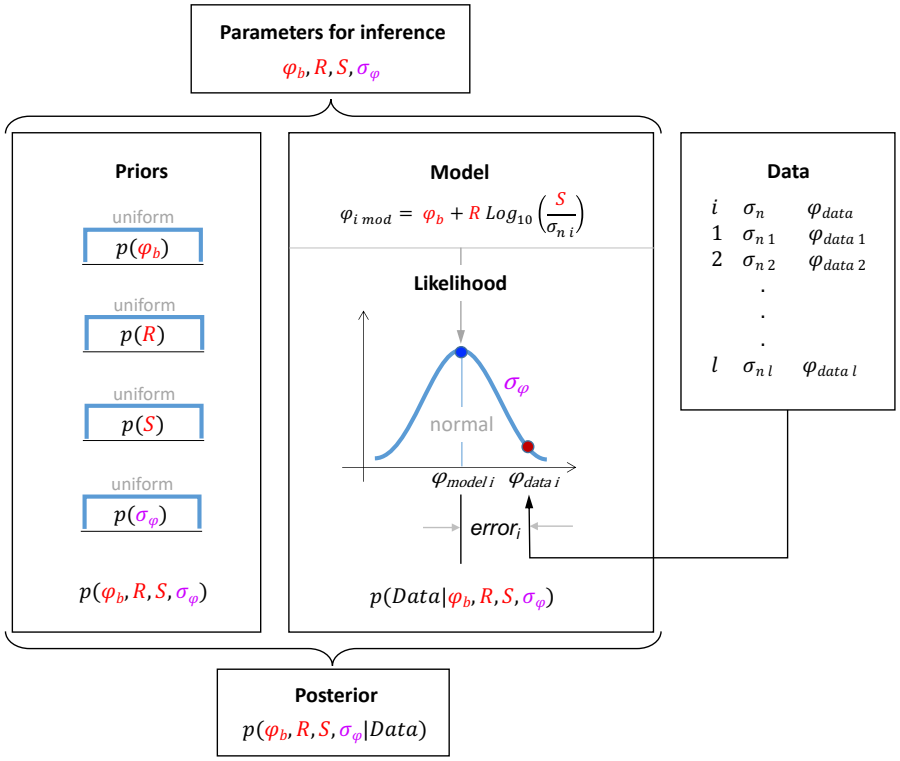


Figure 3: Method for the Bayesian inference of the B-K strength parameters of rockfill material

Figure 4 shows how errors are measured in the Bayesian analysis. A normal distribution is centred in the model prediction and is used to quantify the probability of each data point. In this way, small errors would have larger probability values and vice versa. The product of the probabilities of the data points gives the probability of the dataset for a particular

selection of parameters. The objective of the method is to find the set of parameters that results in large probability values for the dataset, i.e. small errors. For this search, the method should try normal distributions with different widths and for this reason, the standard deviation of the normal distribution used to quantify the errors is an additional parameter that needs to be investigated.

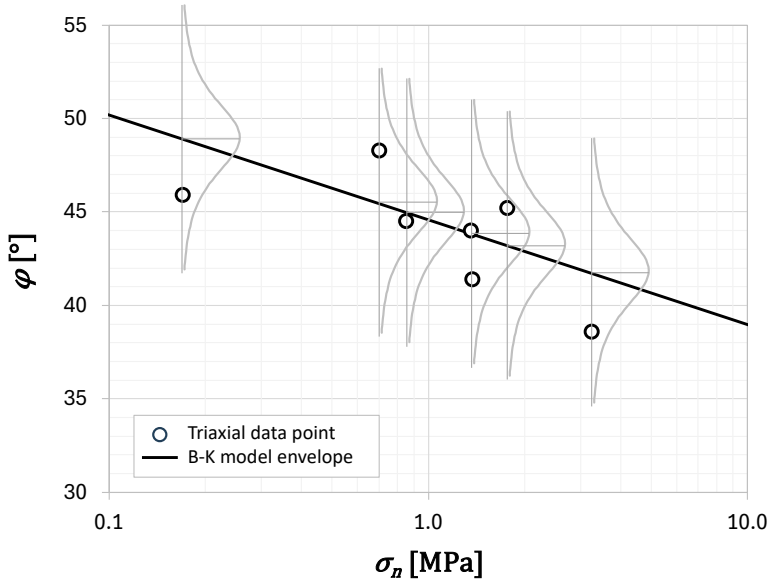


Figure 4: Measurement of errors in φ with a normal distribution centred on the model prediction

The Bayesian model described in Figure 3 can be easily extended to consider the situation where there are two datasets. For example, if there are measurements of basic friction angle from laboratory testing to support the φ_b parameter in the B-K model, the likelihood of this dataset can be evaluated with a normal distribution, similarly as it is described in Figure 3 for the likelihood of the triaxial dataset, but in this case the errors are the difference between the sampled value of φ_b and the measured values in the dataset. The likelihood term in Eq. 8 then becomes the product of the likelihoods of the two datasets.

3.2 BAYESIAN ANALYSIS OF MULTIGROUP DATA

3.2.1 Background

One characteristic of the available dataset of large-scale triaxial tests used to evaluate the shear strength of rockfill materials is that it is limited in the number of tests but covers various planned mixes of different rock types following the requirements for dam construction. Consequently, a conventional overall analysis with no differentiation between material groups treats all the data as belonging to a single group and provides average strength parameters. This type of analysis lacks specificity and fails to identify the individual characteristics of each material group, and thus is not considered in this paper. Instead, the present work focuses on the two additional available options of analysis: the independent analysis of each material group and the hierarchical or multilevel analysis of the dataset, which are conceptually illustrated in Figure 5.

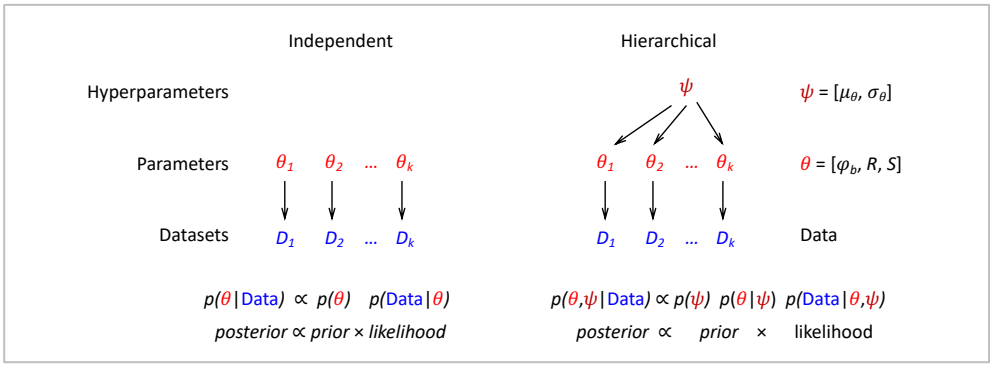


Figure 5: Conceptual illustration of the independent and hierarchical Bayesian models

3.2.2 Independent Bayesian analysis

The independent analysis involves the application of the methodology outlined in Section 3.2 to each subgroup within the dataset, which represents the various rockfill material groups tested. The results derived from this analysis provide specific strength properties for each group; however, they often exhibit a high degree of uncertainty due to the limited number of data points supporting each estimation. In the schematic representation of the independent Bayesian analysis shown in Figure 5, each of the k group analyses is a simplified depiction of the process illustrated in Figure 3. For simplicity, the posterior equation does not include the standard deviation of the errors, σ, as presented in Equation 8.

3.2.3 Hierarchical Bayesian analysis

The primary limitation of the independent analysis can be mitigated by incorporating not only the group-specific data available but also the related information from datasets of other material groups. This can be achieved through hierarchical or multi-level Bayesian analysis. Williams and Ledesma (2022) provide a comprehensive discussion on key features of this methodology and its advantages for analysing geotechnical data from various sources with small sample sizes.

The hierarchical Bayesian analysis considers that while differences exist between the datasets belonging to various material groups, these datasets are not completely unrelated. In the hierarchical framework, there are multiple levels of parameters for the estimation as shown in Figure 5. The top level corresponds to the hyperparameters (ψ), which include the overall means (μ_θ) and standard deviations (σ_θ) of the group parameters (θ). The subsequent level includes the specific material group parameters θ, which include the B-K parameters φ_b, R, S for each material group. During the sampling process of the hierarchical Bayesian model, the distributions of the hyperparameters act as priors for the group-specific parameters. Since the hyperparameters are also unknown, they require priors, which are provided by vague uniform distributions.

The Bayesian hierarchical model represents an intermediate approach between the independent analysis of the various test groups and the single analysis of the complete dataset disregarding group specificity. In the hierarchical analysis all available data support the estimation of the hyperparameters, and each material group's data support the estimation of their respective group-specific parameters. All the information is shared across all the material groups through the link between the hyperparameters and the group-specific parameters. Consequently, as data increase in a specific material group, it receives higher weighting, whereas when data points are few, the prior information influenced by all other material groups has a stronger contribution.

4 CASE STUDY

4.1 BACKGROUND

The case study discussed in this paper is derived from an actual design project. While specific project details are not disclosed due to confidentiality constraints, authentic laboratory testing results were utilized in the analyses. It should be noted that certain aspects of the group configurations, including material proportions and rock types used in the laboratory tests, were adjusted to effectively illustrate the principal contrasting features of the two types of Bayesian analysis described.

4.2 AVAILABLE DATA

This study uses the results of 32 large scale triaxial tests used to investigate the strength characteristics of rockfill materials intended to be used for the construction of tailings dams. The samples tested represent four different material groups constituted by different proportions of five distinct rock types in the project area. To supplement the analysis of the triaxial test results using the B-K strength model, a dedicated campaign of basic friction angle determinations in the laboratory was conducted on samples of the five rock types in the project area that constitute the four rockfill material groups used in the triaxial testing campaign.

The results of the large-scale triaxial tests are presented in Figure 6, along with a general trend line. This trend line is not suitable for design purposes as it fails to differentiate the specific characteristics of the four material groups investigated. This issue is addressed by the two types of analysis discussed in the paper. Table 1 provides a summary of the basic friction angle testing campaign results, while Table 2 outlines the proportions of rock types comprising the four rockfill material groups tested.

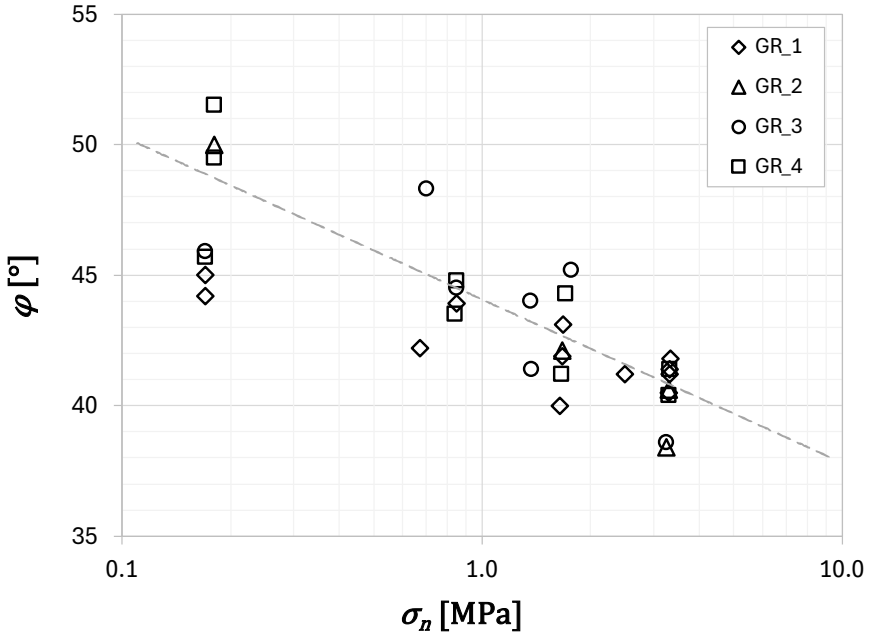


Figure 6: Results of 32 large scale triaxial tests on four material groups used to illustrate the B-K parameter estimation

Table 1: Results of the laboratory determinations of ϕ_b

Rock type	Number of tests	Mean	Standard deviation	Minimum value	Maximum value
RT_1	30	39.5	5.2	26.1	51.7
RT_2	45	38.6	2.3	29.7	42.3
RT_3	30	33.6	3.9	22.5	40.1
RT_4	45	34.9	3.0	27.2	41.4
RT_5	45	32.6	7.8	12.4	43.8

Note: ϕ_b is the basic friction angle from laboratory determinations

Table 2: Percentages of rock types in material groups for the estimation of φ_b

Material groups	Rock types				
	RT_1 (30 ϕ_b tests)	**RT_2** (45 ϕ_b tests)	**RT_3** (30 ϕb tests)	**RT_4** (45 ϕ_b tests)	**RT_5** (45 ϕ_b tests)
GR_1 (12 Triaxial tests)	40%	60%	0%	0%	0%
GR_2 (4 Triaxial tests)	0%	0%	20%	40%	40%
GR_3 (7 Triaxial tests)	12%	18%	14%	28%	28%
GR_4 (9 Triaxial tests)	16%	24%	12%	24%	24%

Note: φ_b is the basic friction angle of a material group from the weighted average of the ϕ_b values of the constituting rock types

4.2 MODEL LAYOUT

4.2.1 Independent Bayesian analysis

A conceptual representation of the model for the independent Bayesian inference of the B-K strength parameters with the case study data is shown in Figure 7. The diagram shows the calculation flow of the posterior probability distribution in two levels from top to bottom. The five grey boxes at the top correspond to the priors of the 13 parameters for inference and the two grey boxes at the bottom represent the likelihood of the two data sets used in the analysis. In this diagram,

the basic friction angle from laboratory determinations for the rock types tested is denoted by ϕ_b, whereas the basic friction angle estimated from the assigned percentages of rock types in the material groups is denoted by φ_b. The posterior probability of a given set of the 13 parameters for inference is calculated as the product of the priors and the likelihoods and the function represented in the diagram can be evaluated with a MCMC procedure to identify the parameter values that minimise the differences between model and data values.

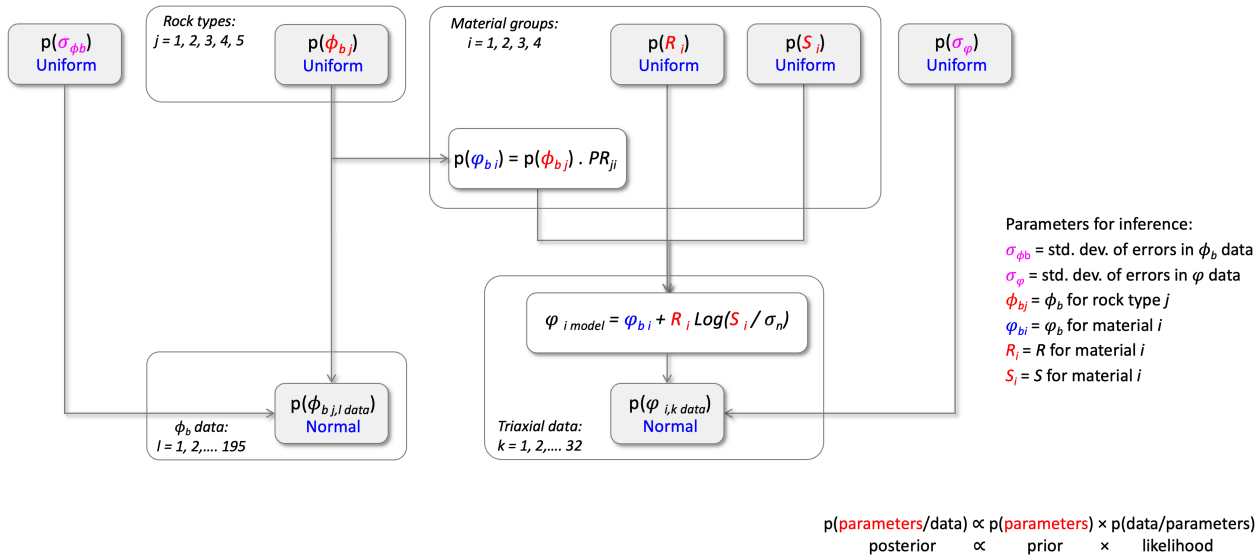


Figure 7: Conceptual representation of the independent Bayesian analysis of data for the inference of B-K parameters

The upper level in the diagram correspond to the priors of the 13 B-K parameters for inference as well as the two priors of the standard deviations of the errors of ϕ_b from laboratory data ($\sigma_{\phi b}$) and φ from triaxial data (σ_φ). The B-K parameters include the rock type-specific ϕ_{bj}, where j=1,2,3,4,5 represents the five rock types tested and the material group-specific R_i and S_i, where i=1,2,3,4 represent the four material groups considered in the analysis. The φ_{bi} prior for the material groups are not sampled directly but calculated as the weighted average of the ϕ_{bj} of the contributing rock types. This is done through the dot product between the vector containing the five ϕ_{bj} values for the five rock types and the [5x4] matrix PR_{ji} containing the percentages of rock types in each material group, summarised in Table 2. The priors are represented by uniform distributions that provide vague information on the parameter values to avoid constraining the inference results. The prior of a parameter x represented with a uniform distribution is calculated from high and low credible values of the parameter as follows:

$$p(x) = \begin{cases} 1/(x_{high} - x_{low}) & , \ x_{low} < \mathrm{x} < \mathrm{x}_{high} \\ 0 & , \ \mathrm{x} < \mathrm{x}_{low} \ or \ x > x_{high} \end{cases} \tag{9}$$

The bottom level in the diagram in Figure 7 corresponds to the probability of the datasets, which is also known as the likelihood of the parameters. In this case, the datasets correspond to 195 values of ϕ_b from laboratory determinations for the five rock types evaluated and 32 values of φ from the triaxial tests across the four material groups tested. The probability of the data is evaluated with a normal distribution centred at the model prediction for each data point, considering the respective standard deviation of the error. The probability of the dataset is equal to the product of the probabilities evaluated for each data point. For the case of ϕ_b data, the model value of the parameter is just the value sampled during the process and the calculation is as follows:

$$p(\varphi_{bj}|data) = \prod_{i=1}^{195} \frac{1}{\sigma_{\varphi b}\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{\varphi_{bj,l}data - \varphi_{bj,l}model}{\sigma_{\varphi b}}\right)^2} \tag{10}$$

Similarly, for the case of the triaxial data, the calculation is as follows:

$$p(\varphi_i|data) = \prod_{k=1}^{32} \frac{1}{\sigma_{\varphi}\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{\varphi_{i,k}data - \varphi_{i,k}model}{\sigma_{\varphi}}\right)^2} \tag{11}$$

The posterior probability of the parameters given the data according to Bayes rule is equal to the probability of the 15 parameters represented in the upper level of the diagram in Figure 7 multiplied by the probability of the two datasets depicted in the bottom level of the diagram.

4.2.2 Hierarchical Bayesian analysis

A conceptual representation of the hierarchical Bayesian model for the inference of the B-K strength parameters with the case study data is shown in Figure 8. The diagram shows the calculation flow in four levels from top to bottom of the posterior probability distribution. The posterior corresponds to the product of the probability of the 21 sampled parameters (represented as coloured variables inside the grey boxes) and two data sets used for the evaluation (grey boxes at the bottom level of the diagram).

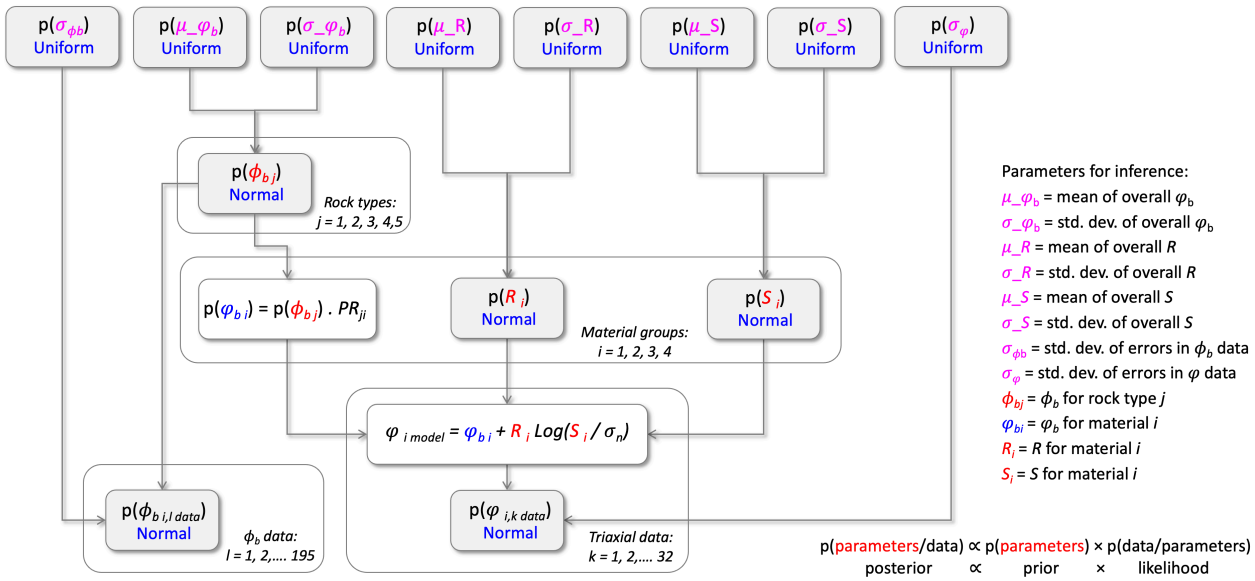


Figure 8: Conceptual representation of the hierarchical Bayesian analysis of data for the inference of B-K parameters

The upper level in the diagram includes the six hyperpriors of the overall B-K parameters and the two priors of the standard deviations of the errors in the two datasets used in the analysis. The hyperpriors correspond to the distributions of the mean (μ) and standard deviation (σ) of the overall φ_b, R and S parameters, which are represented by uniform distributions. The uniform distribution is also used to define the prior of the standard deviations of the errors of ϕ_b from laboratory data ($\sigma_{\phi b}$) and φ from triaxial data (σ_φ). This distribution is selected because it specifies the valid range of the parameter while remaining vague about the exact value within that range, thereby avoiding biasing the results. The prior of a parameter x represented with a uniform distribution is calculated from high and low credible values of the parameter according to Equation 9.

The second and third levels in the diagram correspond to the priors of the B-K parameters for rock types and material groups as applicable. The second level includes the rock type specific ϕ_{bj}, where j=1 to 5 represents the five rock types evaluated and the third level corresponds to the material-specific group parameters R_i and S_i, where i=1 to 4 represent the four material groups considered in the analysis. Similar as for the independent analysis, the φ_{bi} prior for the group materials are not sampled directly but calculated as the weighted average of the ϕ_{bj} of the contributing rock types. This is done through the dot product between the vector containing the five ϕ_{bj} values for the five rock types and the [5x4] matrix PR_{ji} containing the percentages of rock types in each material group, summarised in Table 2.

The 13 priors in levels 2 and 3 of the diagram of Figure 8 (ϕ_{bj} for 5 rock types and R_i, S_i for 4 material groups) are evaluated with normal distributions considering the mean and standard deviation of the respective parent overall parameters in the upper level of the diagram. The prior with a normal distribution of a parameter x can be calculated as follows:

$$p(x) = \frac{1}{\sigma_x\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu_x}{\sigma_x}\right)^2} \quad (12)$$

This way of deriving the priors from the hyperpriors corresponds to a centered parameterization of the random effects of the parameters. However, this often results in zones of high curvature of the posterior distribution that are difficult to

explore with the MCMC sampling procedure. A solution to this drawback is the implementation of a non-centered parameterization, where instead of trying to fit the group-level (material group or rock type) parameters directly, a latent unitary normal distribution ($\boldsymbol{Z} \sim N(0,1)$) is used to recover the group-level parameters with a scaling and a translation (Fonnesbeck et al., 2022). In this case the prior of a group-level parameter x can be calculated as follows:

$$p(x) = \mu_x + \sigma_x \boldsymbol{Z} \tag{13}$$

The fourth level in the diagram in Figure 8 corresponds to the probability of the datasets, which is also known as the likelihood of the parameters. The probability of the datasets of 195 values of ϕ_b for the five rock types evaluated and 32 values of φ from the triaxial tests across the four material groups tested are evaluated in the same way as described for the independent analysis, using Equations 10 and 11, respectively.

The posterior probability of the parameters given the data according to Bayes rule is equal to the probability of the 21 parameters represented in the first three levels of the hierarchical model diagram in Figure 8 multiplied by the probability of the two datasets depicted in the fourth level of the diagram.

4.3 MODEL IMPLEMENTATION

The posterior functions described in the diagrams of Figures 7 and 8 are evaluated with a MCMC procedure to generate representative samples of the distributions. There are various open-source MCMC packages based on different algorithms, with the Metropolis, Gibbs and Hamiltonian algorithms being among the more commonly used. To ensure robust results, MCMC samples must be representative of the posterior distribution, sufficiently large for accurate parameter estimation, and generated efficiently (Kruschke, 2015). Early iterations, known as the burn-in phase, are discarded to stabilize the Markov chain and diagnostic checks, such as acceptance rate or ad hoc evaluations, ensure the quality of the samples.

The models described in this paper were coded in Python using the PyMC library (PyMC, 2022), which is a probabilistic programming language commonly used for Bayesian statistical modelling. The MCMC sampling was carried out with the No-U-Turn sampler (NUTS), which is a variant of Hamiltonian Monte Carlo and the default engine for continuous variables in PyMC. The models were run using 4 chains of 4000 steps each with the initial half of the chains discarded, resulting in 8000 samples to define the inferred parameters. The quality of the samples was verified with the number of divergences diagnostic. The appearance of divergences indicate that the Hamiltonian Markov chain has encountered regions of high curvature in the target distribution which it cannot adequately explore, causing biased estimates (Betancourt et al., 2015). A common way to eliminate divergencies in a hierarchical model is the reparameterization from centered to non-centered approach. The models were implemented using a non-centered parameterization and no divergencies were found in the models evaluated. The quality of the analysis results was also verified with the inspection of diagnostic plots such as the trace plots of the chains and the corner plot of the inferred parameters.

4.4 RESULTS

4.4.1 Independent analysis

The results of the independent analysis of inference of B-K parameters are shown in Figures 9 and 10. Figure 9 shows a corner plot of inferred parameters, which contains scatter plots between the 12 B-K parameters sampled from the posterior distribution and their histograms. The scatter plots illustrate the correlation between the inferred parameters, while the histograms define the ranges of likely values. The plot includes demarcations to facilitate the identification of intra and inter group parameter correlations. These findings indicate a strong correlation between φ_b of groups 2, 3 and 4, which can be attributed to the shared composition of rock types. Additionally, S does not show correlation with any of the other parameters and is constrained by the upper bound of 40 specified for the prior. Figure 9 includes a summary of the mean values of the inferred parameters. However, values within the 95% highest density interval (HDI) of the parameter distributions indicated by the clusters of sampled values in the corner plot are considered acceptable outcomes of the inference analysis.

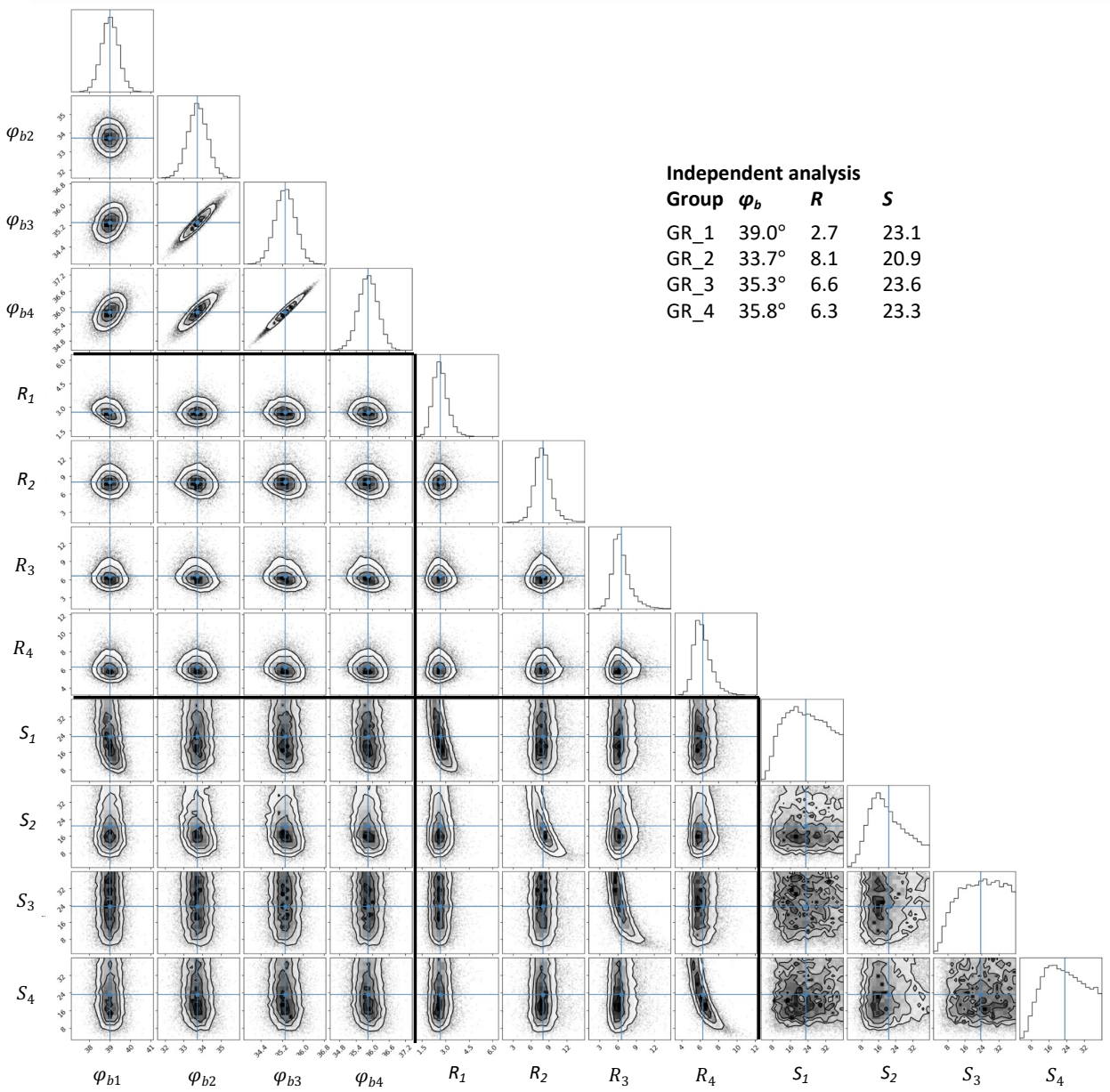


Figure 9: Corner plot of the B-K inferred parameters with the independent analysis of data from material groups

The sampled parameter values depicted in the scatter plots result in a range of B-K envelopes, leading to variability in the estimates of φ, as shown in the graphs of φ versus σ_n in Figure 10, for the four material groups tested. These graphs show the data points, the mean fit envelope, and the band of the 95% HDI of φ derived from the more likely inferred parameter values. The spread of the data points in these graphs determines the width of the 95% HDI of φ.

During the sampling process, a full probability distribution of φ values is constructed at selected σ_n locations to define prediction intervals (PI) of φ with the desired level of confidence. These PIs account for the variability of the mean and standard deviation of the parameters. The graphs in Figure 10 include the band bounded by the 5% and 95% PIs of φ. Figure 11 presents a cross-section of the uncertainty bands for a fixed σ_n value to clarify the interpretation of the 95% HDI and 95% PI of φ. The 95% HDI band shows the uncertainty of the mean value of φ, while the band between the 5% and 95% PIs reflect the uncertainty of φ itself. Critical structures requiring a conservative design are typically evaluated using the lower bound of the PI band corresponding to the 95% confidence level.

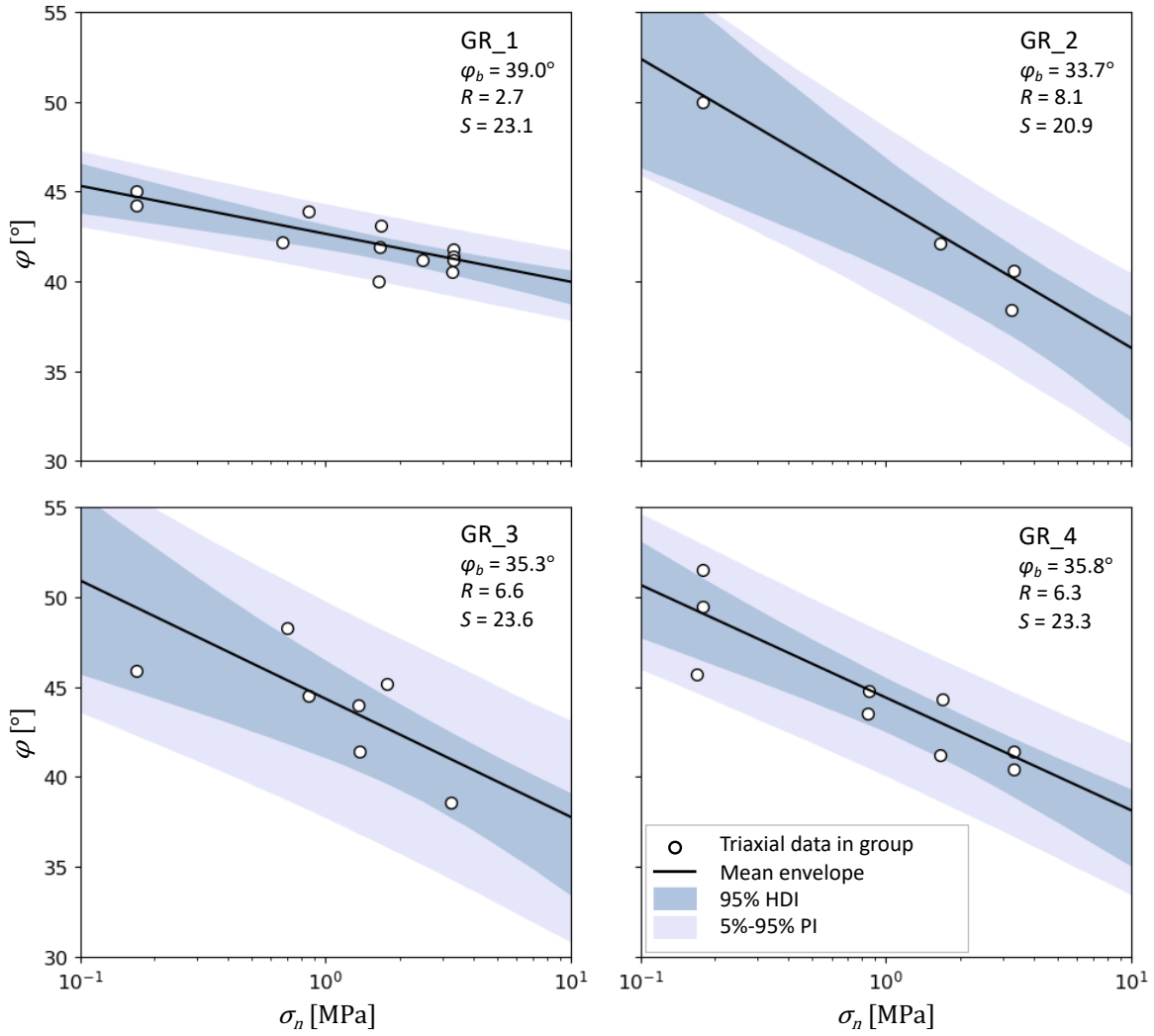


Figure 10: Mean fit and uncertainty bands of φ estimates with the independent analysis of the four material groups

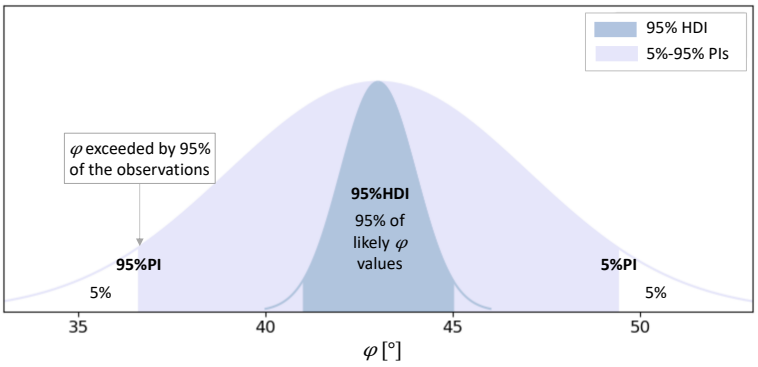


Figure 11: Interpretation of the bands of 95% HDI and 5%-95% PIs representing the uncertainty of φ estimates

4.4.2 Hierarchical analysis

The results of the hierarchical analysis of inference of B-K parameters are shown in Figures 12 and 13. Figure 12 shows a corner plot of inferred parameters, including scatter plots and histograms of the 12 B-K parameters sampled from the posterior distribution. Consistent with the results of the independent analysis, the scatter plots reveal a strong correlation between φ_b of groups 2, 3 and 4, due to the shared composition of rock types among these groups. However, unlike the

independent analysis results, S is not constrained by the prior bounds of the hyperparameter, shows a moderate positive correlation between material groups, and indicates a moderate negative correlation with the parameter R of similar material groups. Figure 12 also includes a summary of the mean values of the inferred parameters.

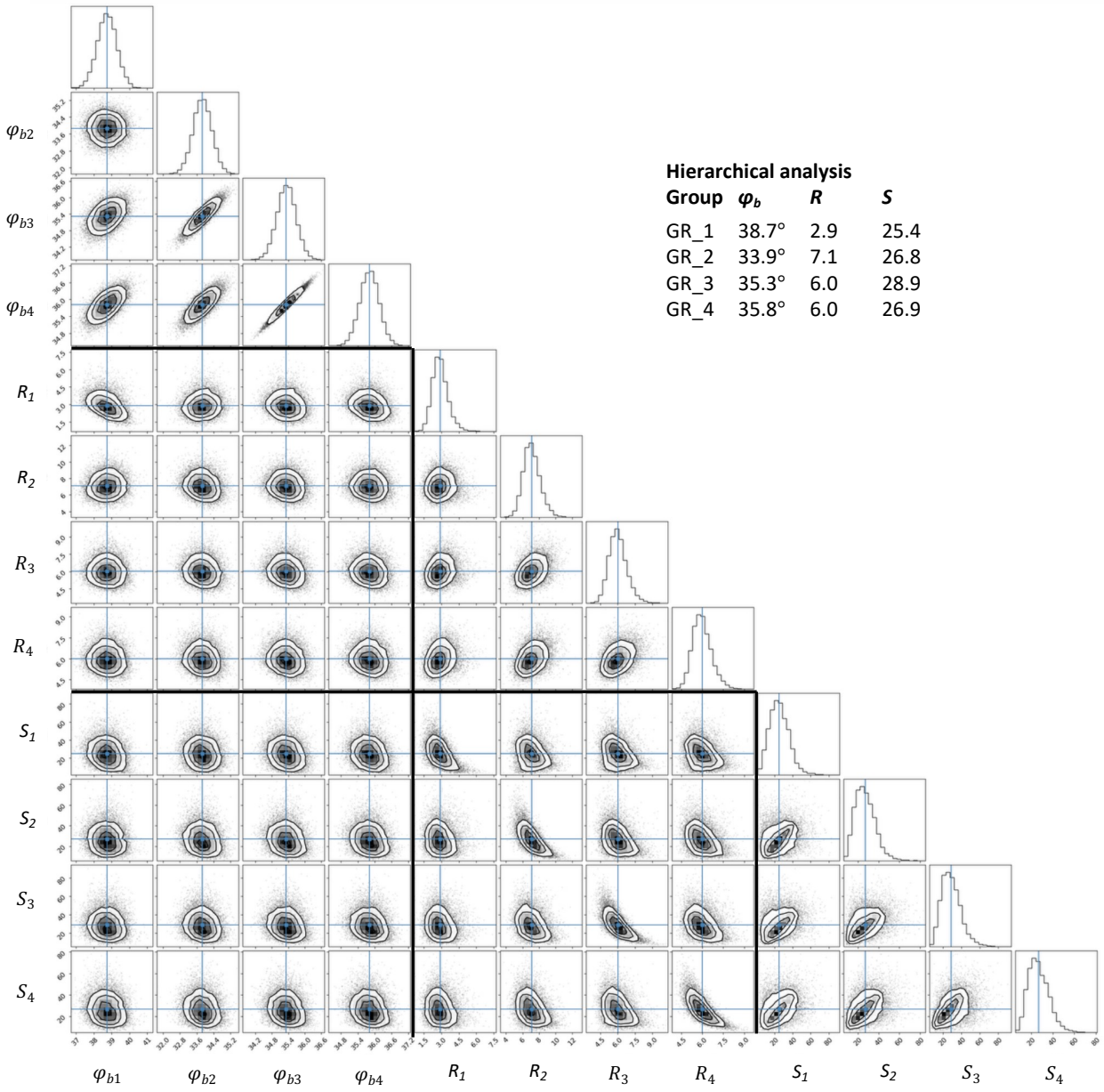


Figure 12: Corner plot of the B-K inferred parameters with the hierarchical analysis of data from material groups

The variability of the estimates of φ is illustrated in the graph of φ versus σ_n shown in Figure 13 for the four material groups tested. The graphs include data points, the mean-fit envelope, and the 95% HDI of the inferred φ values. In contrast with the results of the independent analysis, the narrower 95% HDI bands obtained here, except for group 1, result from the additional information provided by the data points of the other material groups shown in grey in the graphs. This information is transferred through the overall parameters (hyperparameters) in the hierarchical structure of the Bayesian model. For material group 1, the 95% HDI is slightly wider because the 12 data points of this group are distributed along the lower region of the dataset, allowing the wider band to accommodate potential future observations towards the center of the data space. The graphs in Figure 13 also include the band bounded by the 5% and 95% PIs of φ, which aligns with the trend observed for the 95% HDI.

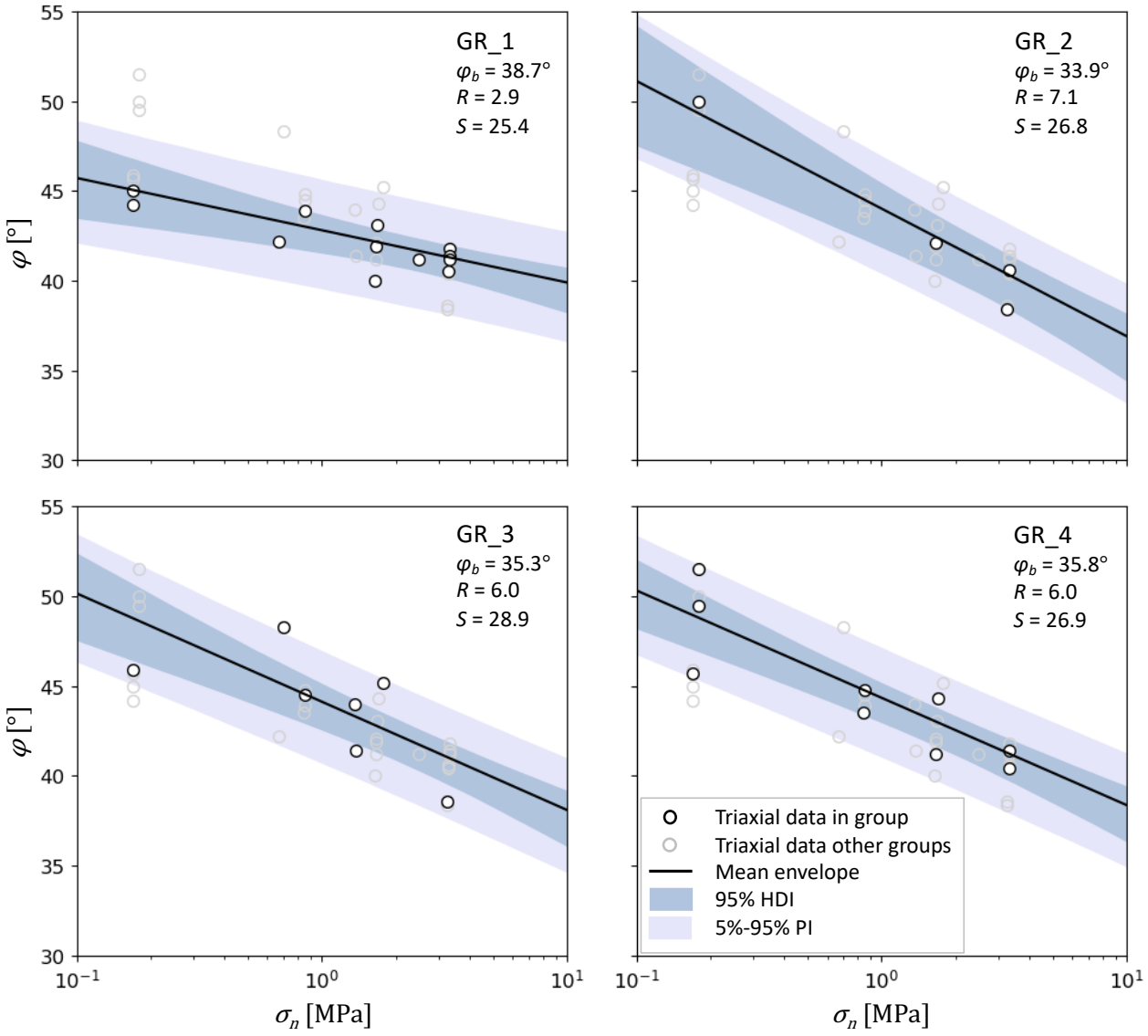


Figure 13: Mean fit and uncertainty bands of φ estimates with the hierarchical analysis of the four material groups

5 DISCUSSION OF RESULTS

Figure 14 shows a comparison between the independent and hierarchical Bayesian estimation results to highlight the main characteristics of the analyses. The most notable observations when comparing the results of the two models are the small changes in the mean fit envelopes and the significant reduction in the variance of φ, except for the material group 1, where the variance moderately increases to compensate for the higher data points of the other groups relative to the material group 1 dataset.

The small variation observed in the mean fit from the hierarchical analysis is attributed to the influence of data points from other material groups, which causes a shift towards the overall mean fit of all 32 data points, effectively acting as the centre of gravity of the dataset. This statistical phenomenon, known as 'shrinkage towards the mean', is transferred to the individual material groups through the hyperparameters that govern the mean and standard deviation of the overall B-K parameters. Despite the broad spread of all data points, the variations in the mean-fit are small because the weight of the specific data points from each material group is considerably higher than the weight of data from other groups.

The changes in the variance of φ and its mean in the hierarchical analysis account for within-group and between-groups variances. The relatively large variance of material group 3, evident in the results of the independent analysis, is significantly reduced when considering the effect of the entire dataset in the hierarchical analysis. Similar results are observed for material groups 2 and 4. However, the relatively small variance in material group 1, supported by a consistent

set of 12 data points, moderately increases in the hierarchical analysis due to the between-groups variances, as material group 1 defines the lowest φ values among all the material groups.

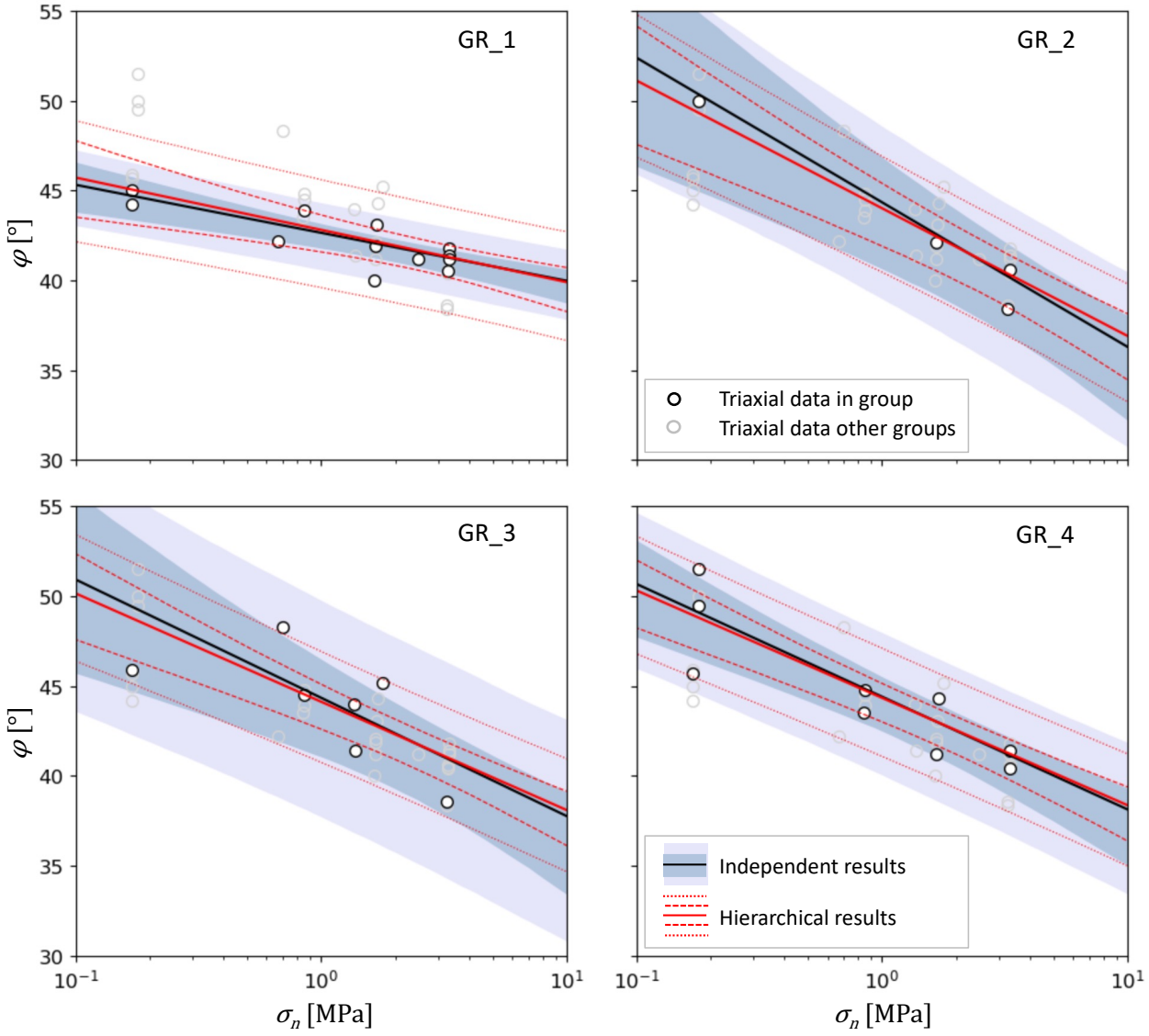


Figure 14: Comparison between the independent and hierarchical model results

The hierarchical analysis method offers an efficient way to leverage available data. It provides a suitable balance between group specific and overall information, considering factors such as the number of data points, mean-fits, and variances at both the material group and overall levels.

6 SUMMARY AND CONCLUSIONS

The Bayesian approach for parameter inference offers several advantages over conventional regression methods, such as providing a formal basis for incorporating various types of information including results from different types of laboratory tests and engineering judgement. Additionally, it offers a robust framework for quantifying uncertainty and can effectively handle small sample sizes. The Bayesian methods discussed in this paper integrate the Barton-Kjærnsli (B-K) strength model with available laboratory data and predefined parameter ranges to create a posterior probability function. This function is evaluated using the Markov Chain Monte Carlo (MCMC) method to estimate the most likely parameter values that minimise the difference between data values and model predictions.

The example case study compares two Bayesian modelling frameworks —independent and hierarchical analyses— to define and quantify the uncertainty of the friction angle of rockfill materials according to the B-K model, which is important for stability analyses of tailings dams. The available test data included results of large scale triaxial compression

tests on different material groups and basic friction angle determinations on different rock types. These methods are necessary when dealing with limited datasets that cover intended mixes of diverse rock types as required for dam construction. A conventional analysis that does not differentiate between material groups lacks specificity and is not suitable.

The results of both type of analysis include scatter plots and histograms of the inferred parameters, along with graphs of φ versus σ_n for each material group. These graphs include bands indicating the uncertainty of φ (5%-95% PI) and its mean (95% HDI). These uncertainty bands are useful as they provide the necessary information for selecting strength values for reliability-based design.

The independent analysis results offer specific strength properties for each group; however, they often exhibit a high degree of uncertainty due to the limited number of data points supporting each estimation. Conversely, in the hierarchical analysis, all available data contributes to the estimation of the overall parameters, while each material group's data support the estimation of their respective group-specific parameters. Information is shared across all the material groups through the connection between the two levels of parameters.

The comparison of the two methods indicated minor shifts of the mean fits and significant adjustments, primarily reductions, in the uncertainty bands with the hierarchical analysis. The hierarchical analysis method uses available data efficiently by balancing group-specific and overall information, considering data points, mean-fits, and variances at both material group and overall levels.

The hierarchical analysis is the recommended approach for multigroup datasets due to its advantages in incorporating all available information, reducing uncertainty, and enabling relative comparisons between groups, compared to the independent analysis. However, the method's applicability depends on whether the groups in the dataset belong to a broader common domain that can be characterised by hyperparameters. Mixing unrelated domains may introduce bias. Therefore, applying this method to a specific dataset requires judgement.

CRediT authorship contribution statement

Luis-Fernando Contreras: Conceptualization, Methodology, Formal analysis, Investigation, Software, Validation, Writing - original draft, Writing - review and editing. **Sandra Linero-Molina**: Formal analysis, Investigation, Validation, Writing - review and editing.

7 REFERENCES

Barton, N. and Kjærnsli, B. (1981). Shear strength of rockfill. *Journal of the Geotechnical Engineering Division*, 107(GT7):873–891.

Betancourt, M. and Girolami, M. (2015). "Hamiltonian Monte Carlo for Hierarchical Models." In *Current Trends in Bayesian Methodology with Applications*, edited by Umesh Singh Dipak K. Dey and A. Loganathan. Chapman & Hall/CRC Press.

Contreras, L.F., Brown, E.T. and Ruest, M. (2018). Bayesian data analysis to quantify the uncertainty of intact rock strength. *Journal of Rock Mechanics and Geotechnical Engineering*, 10(1), 11-31.

Contreras, L.F., Linero-Molina, S. and Dixon, J. (2022). Bayesian approach to improve the confidence of the estimation of the shear strength of coarse mine waste using Barton's empirical criterion. *Australian Geomechanics*, 57(2), 163-174.

Fonnesbeck, C., Carroll, C., Andorra, A., Abril, A. and Reynaldo, F. (2022). A Primer on Bayesian Methods for Multilevel Modelling. In: *PyMC Examples*. Ed. by PyMC Team. DOI: 10.5281/zenodo.5654871.

Hawley, M. (2017). Waste dump and stockpile stability rating and hazard classification system. In *Guidelines for mine waste dump and stockpile design*, M. Hawley and J. Cunning (Eds). CSIRO Publishing.

Kruschke, J.K. (2015). Doing Bayesian Data Analysis: A Tutorial with R, JAGS and STAN, Academic Press/ Elsevier.

Linero-Molina, S., Contreras, L.F. and Dixon, J. (2022). Shear strength estimation of very coarse mine waste using Barton's empirical criterion and a Bayesian analysis. *Proceedings of the 20th International Conference on Soil Mechanics and Geotechnical Engineering,* Rahman and Jaksa (Eds), 4369-4374.

PyMC (2022). PyMC: Probabilistic Programming Library for Python, version 5.16.2 [online]. Available from < https://www.pymc.io/welcome.html > [Accessed: 20 January 2024].

Williams, H. and Ledesma, O.N. (2022). Use of Bayesian Hierarchical models to estimate geotechnical parameters for tailings. *GeoCalgary*, Oct 2-5, 2022.